

Biology
Unit 3
Notes

Question 1

Variables

- Independent: *the one you decide about*
- Dependent: *the one you are measuring*
- Others to be controlled: *the ones that must not change during the experiment*

Tabulation

- * IV comes in first column. Arrange values in ascending order.
- * IV is controlled by the experiment.
- * DV is measured to give the results
- * DV depends on IV
- * Label all columns and rows appropriately and accurately
- * Include SI units in the headings of the columns and rows
- * Be consistent with significant figures/decimal places
- * You are generally asked to find the mean of the percentage difference

Graphs

- * **Axes:** IV on x-axis and DV on y-axis
- * **Scale:** the curve should cover more than 50% of the graph paper
- * **Line:** join each point with a straight line. Do not extrapolate.
- * Describe the trends in the graph. **DO NOT** make theoretical assumptions. Make conclusions based on the experimental data. Refer to the hypothesis being investigated, while making conclusions.
- * Limitations are genuine sources of error. Look out for variables like temperature, pH, etc. not being controlled. Also check if the experiment has been replicated.

Every experiment has two main components:

1. Control of variables/ factors

Ensure that you control all variables and factors that can affect your results. State sensible values, with SI units and describe sensible ways of controlling these variables. Describe the type of apparatus to use.

→ *Controlling factors effectively will improve the reliability of your results*

2. Making quantifiable measurements

The results of your experiment should have numerical values, so that the data can be analyzed and results can be drawn.

→ *Using appropriate method and suitable apparatus can improve the accuracy of your results*

Reliability

- * When the variability in replicated results is very high, the reliability is low. This may be due to certain variables not being controlled or faulty procedures.
- * Range bars or error bars on the graph will give a fair indication of reliability. If there is considerable overlap of between error bars, then variability is very high and reliability is low.
- * Calculating mean and repetition doesn't improve reliability
- * The narrowness of the range of data will tell you if it's reliable or not
- * Results that are close together will have a narrow standard deviation
- * The more variables controlled, the more reliable
- * Repetition by other scientists increases reliability

Accuracy

- * Accuracy can be defined as the difference between the actual values and the measured values. If the difference is high, then accuracy is low and vice versa.
- * Accuracy can be improved by using appropriate apparatus and methods of making measurements
- * Accuracy is given by the gaps between readings in an experiment (small gaps – accurate, big gaps – less accurate)

Technical terms that are commonly used

1. *Precise results* – these are the results taken using sensitive instruments that measure in small increments
2. *Qualitative* – a qualitative test tells you what's present
3. *Quantitative* – a quantitative test tells you how much is present

Question 2

You may be asked to:

1. Say where information should be written within the report
2. Comment on the implications of the Biology in the report
3. Comment on 'bibliographic conventions'
4. Comment on how data is presented

Bibliography

Make sure you state:

1. The exact name of a leaflet/book/journal
2. Give author name
3. Give date
4. Give exact url/address if it's internet
5. Give exact dates of site visits

How to check for validity

1. Look for bias – a sponser – may not be valid
2. Check the contributor – is he really a professor at Harvard?
3. Cross check – look for same info from somewhere else
4. Are there peer views? (proper science papers are looked at by others working in the same field)

Improving answers

“How far does the data support the statements in the passage?”

- Look for differences
- E.g. data says x2 or x3 improvement
But... statement says x20
- Need to say that data does not support passage

Command work “discuss”

- Be prepared to give 2 sides of argument

Comparative questions

- E.g. Rats vs. Daphnia or Drugs vs. Acupunctures
- Use this type of keyword in answer:
“Alternatively”
“on the other hand”
“in contrast”

Effect on caffeine on Daphnia heart rate

- Daphnia lacks physiological methods of maintaining a constant body temperature. This means that if the environmental temperature changes, its body temperature does so too and its metabolic rate will be expected to rise or fall accordingly.
- So body temperature must be kept constant during the procedure.
- Daphnia is relatively transparent and its heart can be seen easily under low Power of the microscope

Setting up the experiment:

1. Make up range of caffeine solutions of different concentrations and a control solution that has no caffeine at all in it
2. Transfer one daphnia into the dimple on a cavity slide
3. Place the slide into the stage of a light microscope and focus on the beating heart of the daphnia
4. Place a small drop of caffeine solution onto the Daphnia
5. Count the number of heartbeats in 10 seconds and multiply this by 6 to calculate beats per minute (heart rate)
6. Repeat this 10 times using the same concentration of caffeine but a different Daphnia individual each time
7. Don't forget to keep all factors constant
8. Repeat experiment using the other concentrations of caffeine
9. Compare the results e.g. draw graph

Ethical issues in the use of invertebrates

1. Daphnia has reduced awareness of pain because of the lack of a well developed nervous system
2. It is transparent and its heart is visible without the need for dissection
3. Daphnia is abundant in nature and there is no threat to it or its dependent species (food chains)
4. Some people feel that it is bred for fish food and will thus die anyway
5. Daphnia can reproduce asexually and may be clone, therefore there is no loss of genetic variation
6. However, they are not given consent and may be subjected to painful procedures
7. It can cause distress and suffering to any living organism (e.g. extreme temperatures)

Independent variable: caffeine concentration

Dependent variable: heart rate of daphnia

Other variables to be controlled:

1. Temperature
2. Volume of solutions
3. Stress of daphnia
4. Size of daphnia
5. Time of acclimatization

6. Habitat from which Daphnia is obtained
7. Oxygen concentration of the water surrounding the daphnia

Possible evaluation issues:

1. Ensuring Daphnia is the same size
2. If left too long under microscope, temperature increases = increased heart rate
3. Too high concentrations can kill daphnia
4. Counting of heart beat can be inaccurate

Measuring the content of Vitamin C in fruit juice

Procedure:

1. Make up several Vitamin C solutions of different known concentrations. Ideally, you need about six different solutions
2. Add Vitamin C solution of a known concentration, drop by drop, to 2 cm³ of the DCPIP (blue) solution in a test tube using a pipette
3. Shake the tube gently after the addition of each drop and continue to add drops until the DCPIP solution is decolorized
4. Record the exact amount of Vitamin C you added
5. Repeat the procedure and calculate the mean volume
6. Repeat the procedure with the fruit juice, containing vitamin C at unknown concentration
7. Record the volume of juice required to decolorize 2 cm³ of the same concentration of DCPIP solution
8. Use results to make a line graph

Note: If only one or two drops of fruit juice are required to decolorize DCPIP, dilute the juice five times and try again

Use the equation below to estimate the concentration of vitamin C in fruit juices

$$\text{concB} = (\text{volA} * \text{concA}) / \text{volB}$$

volA = volume of vitamin C solution in ml

concA = concentration of vitamin C solution in mg ml⁻¹

volB = volume of fruit juice in ml

concB = concentration of vitamin C in fruit juice in mg ml⁻¹

Independent variable: fruit juice

Dependent variable: volume of juice required to decolorize 1 cm³ of DCPIP

Other variables to control:

1. Temperature
2. Concentration of DCPIP solution
3. Shake each tube same number of times
4. Same end point colour

Possible evaluation issues:

1. Difficulty in controlling temperature
2. Amount of shaking (too much adds oxygen will slightly restore the DCPIP to blue)
3. End point difficult to judge
4. Loss of solution when transferring from one beaker to another
5. Accuracy of measuring equipment

The effect of temperature on cell membranes

Procedure:

1. Cut 5 equal pieces of beetroot and rinse them to remove any pigment during cutting (use cork borer to cut)
2. Place the 5 pieces in five different test tubes, each with 5 cm³ of water
3. Place each test tube in a water bath at a different temperature for the same length of time
4. Remove the pieces of beetroot from the tubes, leaving just the coloured liquid
5. Use colorimeter – a machine that passes light through the liquid and measures how much of that light is absorbed. The higher the absorbance, the more pigment is released, so the higher permeability of the membrane

Independent variable: temperature of water

Dependent variable: % transmission of light through resulting solution

Other variables to be controlled:

1. Volume of distilled water
2. Time left in water
3. Size of beetroot piece
4. Storage conditions and age of beetroot
5. Number of beetroot discs
6. Temperature of water bath

Possible evaluation issues:

1. Difficulty in maintaining temperature
2. Accurate reading of the colorimeter
3. Accurate size of beetroot
4. From the different parts of the root
5. Ensuring the same time at the different temperatures

The effect of changing enzyme concentration on rate of reaction

Safety: wear eye protection and protect clothing from hydrogen peroxide

Equipment:

1. Upside down measuring cylinder
2. Delivery tube
3. Solution of hydrogen peroxide and catalase

Procedure:

1. Set up experiments with equipment above – it shows hydrogen peroxide being broken down into water and oxygen by the enzyme catalase
2. You can measure how much oxygen is given off with different concentrations of enzyme
3. Measure the amount of oxygen given off in the first minute of the reaction
4. Divide the volume of oxygen produced by time taken, to get the initial rate of reaction (in cm^3/min)
5. The more oxygen that is given off in the first minute of the reaction, the faster the initial rate of reaction

Independent variable: concentration of enzyme

Dependent variable: time taken for enzyme to break down substrate

Other variables to control:

1. Temperature
2. Volume of enzyme
3. Volume of substrate
4. Concentration of substrate
5. pH

Possible evaluation issues:

1. Maintaining constant temperature
2. Attaching syringe
3. Inaccurate readings
4. Reaction too quickly to read

Observing mitosis

Method:

1. Cut the tip from a growing root (e.g. broad bean). Should be about 5mm long.
2. Place root tip on a watch glass and a few drops of hydrochloric acid
3. Add a few drops of stain so that the chromosomes become darker and so easier to see under a microscope. (Acetic orcein)
4. Warm the watch glass (but don't boil the liquid) by passing it slowly through a Bunsen burner flame
5. Place root tip on a microscope slide and use a mounted needle to break it open and spread the cells out thinly
6. Add a few more drops of stain and then place a cover slip over it
7. Squash the cover slip down gently
8. Warm the slide again for a few seconds. This will intensify the stain.
9. Now you can look at all the stages of mitosis under a light microscope

Possible evaluation issues:

1. Resolution of microscope
2. Human error in counting numbers of cells
3. Enough time in the solutions to enable successful maceration or staining

Totipotency and tissue culture

- Plants have stem cells – found in areas of growth (e.g. roots or shoots)
- All stem cells in plants are totipotent
- Totipotency can be shown in plants using tissue culture, method used to grow a plant from a single cell.

Method:

1. A single cell is taken from a growing area on a plant
2. The cell is placed in some growth medium (e.g. agar) that contains nutrients and growth hormones. The growth medium is sterile, so microorganisms can't grow and compete with plant cells
3. The plant cell will grow and divide into a mass of unspecialized cells. If the conditions are suitable the unspecialized cells will differentiate into specialized cells
4. Eventually, the cells will grow and differentiate into an entire plant

Possible evaluation issues:

1. Unwanted pathogens growing in the gel as it is a good source of water and nutrients
2. Wrong part of the plant cut and inserted into the gel

The strength of plant fibres

→ Tensile strength – maximum load the fibre can take before it breaks

Method:

1. Attach the fibre to a clamp stand and hang a weight from the other end
2. Keep adding weights, one at a time, until the fibre breaks
3. Record the mass needed to break the fibre – the higher the mass, the higher the tensile strength
4. Repeat the experiment with different samples of same fibre – increases reliability
5. The fibres being tested should always have the same length
6. All variables must be kept constant – e.g. temperature, humidity
7. You also need to take safety measures when doing this experiment, e.g. wear goggles to protect your eyes and leave the area where the weights will fall clear

Independent variable: source and type of fibre

Dependent variable: mass that can be held

Other variables to be controlled:

1. Length of fibre
2. Size of each individual mass
3. Temperature
4. Humidity

Possible evaluation issues:

1. Maintaining lengths of fibres
2. Ensuring consistency when twisting or plaiting
3. Using fibres of same age (as they get older, they become brittle)
4. Extracting whole fibres that are useful

Investigating plant mineral deficiencies

Method:

1. Take 30 seedlings of the same plant (they should be the same age and height) and plant them in separate pots
2. Make up three nutrients broths (definition: liquid medium containing proteins and other nutrients for the culture of bacteria) containing all essential minerals, but vary the concentration of calcium ions. Make up one broth with a high concentration, one with medium and one with low concentration of calcium ions.
3. Split the plants into three groups. Each group should be given only of the three broths.
4. Record the heights of the plants after several weeks. Calculate average height of each group
5. During the experiment, it is important to keep all other variables the same
6. The greater the concentration of calcium, the more plants grow – this shows that when calcium is deficient – the more the plants grow

Independent variable: concentration of mineral

Dependent variable: physical characteristics of the plant (height in this case)

Other variables to control:

1. Volume of mineral solution
2. Species of plant
3. Size of container
4. Amount of light received
5. Amount of water received

Possible evaluation issues:

1. Ensuring accurate measurements of solutions
2. No air bubbles caught in xylem of geranium
3. Possible microorganisms growth in nutrient solution
4. Insufficient time to see an effect

Effect of garlic and mint on bacterial growth

- Some plants have antimicrobial properties – they kill or inhibit the growth of microorganisms

Method:

1. Take extracts from the plant you want to test. To do this you need to dry and grind each plant, then soak them in ethanol (acts as solvent). The plants should all be the same size, so the amount of extract is the same.
2. Filter off the liquid bit (the ethanol containing the dissolved plant extract)
3. You need some bacteria to test the plant extract on – evenly spread a sample of bacteria onto an agar plate
4. Dip discs of absorbent paper in the extract. The discs of paper should all be the same size so they absorb the same volume of liquid
5. You also need to do a control disc soaked only in ethanol (to make sure it isn't the ethanol or the paper that's inhibiting bacterial growth)
6. Place the paper discs on the agar plate – make sure they're spread out
7. Incubate the plate to allow the growth of bacteria
8. When bacteria can't grow there'll be a clear patch in the lawn of bacteria. This is called an inhibition zone.
9. The size of an inhibition zone tells you how well the antimicrobial plant extract is working. The larger the zone, the more effective the plant extract is.

Independent variable: **presence of garlic or mint**

Dependent variable: **zone of inhibition around the disc**

Other variables to be controlled:

1. Concentration of plant material
2. Lawn of bacteria on petri dish
3. Contamination of petri dish by other microbes
4. Same volume of plant material on each disc

Possible evaluation issues:

1. Growth of unwanted microbes on agar plates due to bad aseptic techniques
2. Not shaking extract enough to ensure enough active ingredient
3. Inconsistency when adding plant extract to paper discs
4. Contaminating controls
5. Using wrong species of bacteria for lawn